

S PREETHAM

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SUMMARY

Software Engineer with experience building AI-powered products, scalable backend systems, and modern full-stack applications. Skilled in Python, Java, TypeScript, FastAPI, React, and Next.js, with hands-on expertise in LLMs, RAG, vector databases, REST APIs, and cloud-native development. Strong computer science fundamentals and a proven ability to deliver end-to-end software solutions from design to deployment.

TECHNICAL SKILLS

Languages: Python, Java, TypeScript, JavaScript

Backend: FastAPI, REST APIs, Microservices, Node.js, Express.js, gRPC

Frontend: React.js, Next.js (SSR/SSG), Tailwind CSS, Zustand, HTML5, CSS3, Accessibility

Databases: PostgreSQL, MongoDB, Redis, ChromaDB (Vector DB)

AI / ML: RAG, LangChain, LLM Evaluation, LLM Fine-tuning, Agentic AI ASR/TTS, Machine Translation, Hugging Face, Scikit-learn, Pytorch, MLflow, Vector Embeddings

DevOps / Tools: Docker, GitHub Actions (CI/CD), Git, Postman, Linux, Chrome DevTools Protocol, Vercel, Render, GCP

CS Fundamentals: DSA, OOP, DBMS, Operating Systems, Computer Networks, System Design

EXPERIENCE

ML Engineer Intern

Jan 2026 – May 2026

Assisto Technologies Pvt Ltd · Noida

- Architected cdpdriver, a custom browser automation framework using Chrome DevTools Protocol (CDP), reducing container memory overhead by 30% replacing Selenium/Playwright with a lighter, scriptable solution deployed via Docker.
- Designed and deployed an ML-based query-routing microservice using REST APIs, mapping user queries to knowledge nodes with vector embeddings, improving retrieval precision for a conversational AI product, reducing hallucination rates by 14%.
- Fine-tuned ASR and pretrained models (Whisper / Hugging Face) and Neural Machine Translation models for 30 languages; integrated MLflow for experiment tracking and model versioning.
- Built a post-call analytics pipeline extracting sentiment, intent, and action items from conversational data using LLM-based evaluation; results surfaced on a real-time dashboard.

Web Development Apprentice

Oct 2023 – Nov 2023

Lam House · Bengaluru

- Redesigned 15+ webpages into fully responsive experience using CSS Grid and Flexbox, resulting in a 20% increase in mobile user engagement metrics.
- Improved performance across 20+ media assets through image optimization, lazy loading, and reducing render-blocking resources, resulting in faster page load times.
- Enhanced UI accessibility (WCAG standards) by adding semantic HTML, ARIA labels, and keyboard navigation support, improving usability for a wider audience.
- Collaborated with stakeholders to gather requirements, iterated on designs based on feedback, and delivered the project within a 1-month timeline.

PROJECTS

TRIACT - AI-Powered Inventory & Stock Management System

[Github](#) | Jun 2025 – Nov 2025

FastAPI · PostgreSQL · React · RAG · LangChain · Vector DB (ChromaDB) · Docker · GitHub Actions

- Built a multi-tenant SaaS app with stateless FastAPI services and PostgreSQL connection pooling to handle concurrent tenant queries; enforced strict row-level data isolation.
- Engineered a per-tenant RAG pipeline using LangChain and ChromaDB (vector database) to answer natural-language queries over invoices, sales history, and stock data.
- Automated invoice processing by implementing a CI/CD pipeline with GitHub Actions, reducing manual data entry time by 90% through OCR and structured extraction.

Enterprise Virtual Interview & Event Platform

[Github](#) | Feb 2025 – Jun 2025

Next.js 13 (SSR/SSG) · TypeScript · Convex · Clerk · LiveKit · Zustand · Tailwind CSS

- Developed a full-stack events platform with SSR/SSG via Next.js 13, Clerk auth, Zustand state management, real-time chat, Q&A, ticketing, and LiveKit video streaming.
- Implemented Row-Level Security (RLS) in Convex for user-level access control; built accessible, reusable component library improving engagement across host and attendee dashboards.

EDUCATION

B.E. Computer Science & Engineering - Don Bosco Institute of Technology, Bengaluru

Graduated Jun 2026 | CGPA: 8.47 / 10

Coursework: Data Structures & Algorithms, OOP (Java), Machine Learning, DBMS, Operating Systems, Computer Networks

CERTIFICATIONS

MACHINE LEARNING, NPTEL

Jan 2025 -- Mar 2025

JAVA FOUNDATION CERTIFICATION, Infosys SpringBoard

May 2025 – Jun 2025

ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING FOUNDATION, TNS India Foundation

Aug 2025 – Oct 2025